Drift-Regularized Multi-Token Prediction via On-Policy Self-Distillation

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1 Objective

This research investigates latent representation alignment in policy distillation, aiming to improve multi-token generation beyond supervision based solely on token-level predictions. While conventional knowledge distillation primarily constrains output distributions, the geometry of intermediate latent representations remains underdetermined.

Our goal is to establish how latent geometric alignment can improve the accuracy–efficiency trade-off in multi-token prediction. To this end, we develop a geometry-aware drifting regularizer that aligns student and teacher latent-state distributions, and compare it with conventional token-level supervision and paired feature regression. We will like to demonstrate that population-level drifting offers advantages over pointwise regression and token-level alignment in terms of better thoughput and latency while maintaining or improving reasoning accuracy.

2 Novelty

Our contributions are twofold: introducing drifting regularization [2] as a latent-space supervision objective for multi-token prediction (MTP) self-distillation, and systematically investigating how its attraction–repulsion components, sampling strategies, and hyperparameters influence representation alignment and generation performance.

First, we adapt drifting to language-model hidden representations, where teacher states serve as positive samples and other student states form the negative population. The resulting attraction–repulsion field encourages population-level alignment without enforcing exact pointwise correspondence, providing an alternative to conventional paired feature regression (Section 4).

Second, we decompose the drifting formulation into three components to examine their respective contributions and identify suitable sampling strategies (Section 4.2). We further investigate how these design choices affect the alignment of teacher and student feature distributions and their relationship to multi-token rollout performance.

3 Impact

We hypothesize that improved latent representation alignment can enhance multi-token prediction beyond the gains achievable through token-level self-distillation alone. In particular, preserving the teacher's representation geometry may help the student maintain coherent and distinguishable predictive states across longer prediction horizons, potentially improving generation quality without sacrificing parallelism.

The proposed approach could enable the model to produce more reliable tokens per forward pass, thereby reducing generation latency and increasing token throughput under standalone MTP inference. Since drifting regularization is applied only during training, it introduces no additional inference module or verification stage.

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More broadly, this study aims to clarify the relationship between representation geometry, token-level prediction quality, and the accuracy–acceleration trade-off in self-distilled language models. Such findings could inform the design of representation-level supervision for other parallel generation and model distillation frameworks.

4 Methods

We introduce the notation as needed throughout this section. For convenience, a consolidated summary of the symbols and their definitions is provided in Appendix 7.

4.1 On-policy MTP self-distillation

We adopt the multi-token prediction (MTP) framework of Kirchenbauer et al. [5], which converts a pretrained autoregressive language model into a standalone multi-token predictor through on-policy self-distillation. The pretrained model serves as a frozen teacher, while a student initialized from the same checkpoint learns to predict multiple future tokens simultaneously. The teacher evaluates the student's proposed continuation autoregressively, providing token-level supervision without requiring an additional verification model during inference.

During the training, for each region r, a prefix c_r is followed by $k_r - 1$ trainable `<MTP>` mask tokens, allowing the student to propose k_r future tokens in one forward pass. Repeated position IDs and blocked causal attention preserve the original context computation while allowing each prediction state to attend to the prefix and preceding masks. At horizon j, the student produces latent state $u^S_{r,j}$ and distribution $p^S_{r,j}$, yielding a greedy proposal $\hat{y}_{r,j} = \arg\max_v p^S_{r,j}(v)$.

The teacher then supplies targets on the continuation actually proposed by the student, rather than on a fixed ground-truth continuation. With a frozen autoregressive teacher,

$$u^T_{r,j} = H_{\theta_T}(c_r, \hat{y}_{r,<j}), \qquad \bar{y}_{r,j} = \arg\max_v p_{\theta_T}(v \mid c_r, \hat{y}_{r,<j}). \tag{1}$$

The hard-teacher token objective is

$$\mathcal{L}_{\mathrm{MTP}} = -\frac{1}{N_{\mathrm{act}}} \sum_{r,j} \log p^S_{r,j}(\bar{y}_{r,j}), \qquad N_{\mathrm{act}} = \sum_r k_r. \tag{2}$$

This common token objective is retained in every experimental family so that changes in rollout behaviour can be attributed more clearly to the added representation losses.

4.2 Latent-space supervision

The latent representations of student and teacher prediction tokens are paired at corresponding prediction positions. We introduce SmoothL1 regression as one of the baselines to align each student latent state with its corresponding teacher state, providing direct representation-level supervision to complement the hard-teacher token loss used in MTP training. For active position $i = (r, j)$, let $x_i = \phi_S(u^S_i)$ and $y^+_i = \phi_T(u^T_i)$ denote student and frozen-teacher features under a shared normalization. Paired SmoothL1 regression supplies a direct target for each student state:

$$\mathcal{L}_{\mathrm{SmoothL1}} = \frac{1}{N_{\mathrm{act}} d} \sum_{i=1}^{N_{\mathrm{act}}} \sum_{m=1}^{d} \rho_\beta\big(x_{i,m} - \mathrm{sg}(y^+_{i,m})\big), \tag{3}$$

where $\rho_\beta(z) = z^2/(2\beta)$ for $|z| < \beta$ and $|z| - \beta/2$ otherwise. This baseline evaluates the benefit of pointwise teacher–student alignment, although enforcing individual correspondences may restrict the flexibility of the student representation distribution.

In contrast, drifting constructs a population-level update from interactions between neighbouring teacher and student representations. Drifting regularisation allows the student to approximate the teacher's representation geometry without explicitly enforcing pointwise agreement between corresponding latent states. Following Deng et al. [2], we use a feature-space Laplace affinity for prediction tokens:

$$\kappa_h(x_i, y_l) = \exp\left(-\frac{\|x_i - y_l\|_2}{\tau_h}\right). \tag{4}$$

Here, x_i denotes the student latent-state feature at prediction position i, serving as the anchor for the drifting field. The positive pool $\mathcal{J}_i^+$ consists of teacher latent-state features y_l^+, including the teacher state paired with x_i (self-positive configuration). The negative pool $\mathcal{J}_i^-$ consists of other student latent-state features y_l^-, excluding the anchor itself ($l \neq i$). Both pools provide feature samples against which the anchor is compared, with teacher states contributing attraction and other student states contributing repulsion. For $s \in \{+, -\}$, the within-pool affinities are

$$a_{il}^s = \frac{\kappa_h(x_i, y_l^s)}{\sum_{m \in \mathcal{J}_i^s} \kappa_h(x_i, y_m^s) + \epsilon}, \qquad l \in \mathcal{J}_i^s, \tag{5}$$

with $y_l^- = \mathrm{sg}(x_l)$. Teacher neighbours attract the student state while student neighbours supply a repulsive term:

$$\widehat{V}_i = \sum_{l \in \mathcal{J}_i^+} a_{il}^+ (y_l^+ - x_i) - \sum_{l \in \mathcal{J}_i^-} a_{il}^- (y_l^- - x_i). \tag{6}$$

We train against a stopped, drifted target,

$$\mathcal{L}_{\mathrm{drift}} = \frac{1}{N_{\mathrm{act}}} \sum_i \left\| x_i - \mathrm{sg}\left(x_i + \eta \frac{\widehat{V}_i}{\|\widehat{V}_i\|_2 + \epsilon} \right) \right\|_2^2. \tag{7}$$

Here η controls the magnitude of the drift-induced target displacement. Pool composition and horizon grouping are implementation choices, separate from the kernel. Unlike paired regression, this formulation encourages the student representation distribution to approach that of the teacher without requiring exact correspondence between individual latent states.

Unlike the original drifting formulation for generative modelling, on-policy MTP self-distillation provides an explicit correspondence between each student prediction state x_i and its teacher counterpart y_i^+. Although this correspondence provides additional structural information, it also introduces challenges in constructing balanced positive and negative populations for the drifting field, as discussed in the following section.

Paired self-distillation and self-negative correction The construction of the positive and negative pools $\mathcal{J}_i^+$ and $\mathcal{J}_i^-$ requires careful consideration of sample selection. In particular, including the student anchor itself in the negative pool or retaining its paired teacher representation in the positive pool can introduce undesirable weighting effects. We therefore introduce a correction to mitigate these issues which will be introduced at the end of this section.

Directly including the student anchor x_i in the negative population can cause its self-affinity $\kappa_h(x_i, x_i) = 1$ to dominate the normalized negative weights, despite producing zero displacement. Conversely, including the paired teacher y_i^+ in the positive population while excluding x_i from the negative population introduces an asymmetric weighting of paired and unpaired interactions.

To characterize this asymmetry, we define the normalized paired-teacher affinity a_i, the weighted mean P_i of other teacher features, and the weighted mean N_i of other student features:

$$\begin{aligned} a_i &= a_{ii}^+, \\ P_i &= \frac{1}{1 - a_i} \sum_{j \neq i} a_{ij}^+ y_j^+, \\ N_i &= \sum_{j \neq i} a_{ij}^- x_j. \end{aligned} \tag{8}$$

Here, a_{ij}^+ and a_{ij}^- are the row-normalized teacher and student affinities, respectively, with the student diagonal excluded from a_{ij}^-. The definition of P_i applies when $a_i < 1$; otherwise its weighted contribution vanishes.

Table 1: Training objectives and experimental configurations. M0 provides the token-only baseline, M1 and M2 compare population-level drifting with pointwise feature regression, and M3 evaluates the tied self-negative correction described in Section 4.2. The legend labels are used consistently across figures.

ID	Figure legend	Training objective	Purpose
M0	`MTPloss`	Hard-teacher MTP only	Token-supervision baseline
M1	`Drift+MTPloss`	MTP + diagonal-masked drifting	Population-level alignment baseline
M2	`SmoothL1+MTPloss`	MTP + paired SmoothL1	Pointwise feature-regression baseline
M3	`TieDrift+MTPloss`	MTP + tied self-negative drifting	Self-negative correction ablation (Section 4.2)

The resulting drifting field admits a three-term decomposition:

$$\begin{aligned}\widehat{V}_i^{\mathrm{masked}} = & \underbrace{a_i(y_i^+ - x_i)}_{\text{paired attraction}} \\ & + \underbrace{(1-a_i)(P_i - N_i)}_{\text{population correction}} \\ & + \underbrace{a_i(x_i - N_i)}_{\text{residual repulsion}}.\end{aligned} \tag{9}$$

The first term attracts the student towards its paired teacher, while the second captures the difference between the remaining teacher and student populations. The third term is an additional repulsive contribution caused by the mismatch in normalization: unpaired teacher features carry total weight $1 - a_i$, whereas other student features retain unit total weight.

To eliminate this residual term, we introduce a *tied self-negative* construction that assigns the student anchor the same negative weight as its paired teacher's positive weight:

$$\widetilde{a}_{ij}^- = \begin{cases} a_i, & j = i, \\ (1-a_i)a_{ij}^-, & j \neq i. \end{cases} \tag{10}$$

This preserves the normalization $\sum_j \widetilde{a}_{ij}^- = 1$ while maintaining the relative affinities among other student features. The resulting field becomes

$$\boxed{\widehat{V}_i^{\mathrm{tied}} = a_i(y_i^+ - x_i) + (1-a_i)(P_i - N_i).} \tag{11}$$

Although the self-negative contributes no direct displacement, its assigned weight reduces the repulsive contribution from other student states. Under exact pointwise teacher–student matching, the tied field vanishes, unlike the diagonal-masked formulation.

We compare drifting with the student anchor excluded from the negative population against the proposed tied self-negative construction. This ablation investigates whether eliminating the residual repulsive term improves representation alignment and downstream multi-token prediction performance.

4.3 Experimental design: drifting and its baselines

The experiments are designed to distinguish the effects of population-level alignment, pointwise feature regression, and self-negative correction. M0 provides the token-only reference, while M1 and M2 compare diagonal-masked drifting against paired SmoothL1 regression. M3 introduces the tied self-negative construction described in Section 4.3, isolating the effect of correcting the asymmetric attraction–repulsion weighting. All configurations retain the same hard-teacher MTP objective to enable controlled comparisons.

We will compare these families at matched initialization, data exposure, and evaluation settings wherever feasible. In the pilot configurations, the drifting coefficient $w_{\mathrm{drift}} = 0.05$ corresponds to an approximately 30% auxiliary-to-token gradient-norm ratio; SmoothL1 with $w_{\mathrm{SmoothL1}} = 0.53$ corresponds to approximately 20%.

4.4 Evaluation: from feature geometry to rollout performance

A geometric improvement matters for this proposal only if its relationship to generation can be measured. We therefore separate latent diagnostics from end-to-end rollout outcomes. The current pilot uses MetaMathQA for training and GSM8K for free-rollout evaluation; broader benchmarks will be chosen for larger checkpoints. The frozen autoregressive teacher under sequential decoding ($k = 1$) is the capability-retention reference.

Feature spread, maximum mean discrepancy (MMD) [3], and Sinkhorn transport cost [1] assess different aspects of teacher–student geometry.We also evaluate token-level expected calibration error (ECE) [4] to assess how closely the model's predicted token confidence reflects its empirical prediction accuracy. All metrics are reported separately for each prediction horizon to isolate horizon-dependent effects and avoid confounding trends arising from aggregating predictions across different horizons. Free-rollout accuracy is compared against the effective number of emitted tokens per decoding pass,

$$k_{\text{eff}} = \frac{\text{emitted non-padding tokens}}{\text{decoding forward passes}}. \tag{12}$$

Both fixed-k and confidence-adaptive decoding will be assessed. Measured tokens per second and end-to-end latency will be reported separately, because k_{eff} is a proxy for acceleration rather than a wall-clock speedup.

4.5 Preliminary study and current progress (October 2026)

The 1B pilot investigates how different forms of latent supervision affect teacher–student representation geometry and downstream multi-token generation. In particular, we examine whether the tied self-negative correction improves the alignment properties of conventional drifting while preserving its population-level geometric behaviour. The on-policy MTP reproduction is complete; the current results are exploratory single-run comparisons rather than controlled multi-seed evaluations. The four experimental configurations are summarized in Table 1.

Geometric spread: drifting preserves relative dispersion. Figure 1 compares the ratio of mean pairwise student and teacher feature distances at prediction horizon $h = 3$. Paired SmoothL1 regression produces the strongest contraction of student representations, whereas conventional drifting maintains a spread ratio close to unity. The tied self-negative formulation exhibits similarly preserved feature dispersion, suggesting that correcting the asymmetric repulsive field does not substantially compromise this property of drifting. However, agreement in average pairwise distance alone does not establish complete distributional alignment.

Distribution alignment: tied drifting improves short-horizon matching. Figure 2 evaluates teacher–student alignment using maximum mean discrepancy (MMD) and Sinkhorn transport cost across prediction horizons. Conventional drifting generally improves MMD relative to token-only MTP at longer horizons, but exhibits weaker alignment at the first prediction position. The tied self-negative formulation reduces this short-horizon discrepancy, particularly at $k = 1$, while maintaining comparable MMD at longer horizons. This behaviour is consistent with the proposed self-negative correction: excluding the student anchor while retaining its paired teacher introduces an additional repulsive contribution $a_i(x_i - N_i)$, which may displace student features away from their paired teacher representations. Removing this residual term may therefore be particularly beneficial at short horizons, where the prediction state more closely resembles the next-token prediction (NTP) setting. Nevertheless, the Sinkhorn results show that SmoothL1 remains competitive, indicating that different regularizers favour distinct aspects of representation alignment.

Rollout performance: geometric alignment and generation accuracy. The practical objective is to improve generation accuracy at a given effective number of emitted tokens per pass. Figure 3 compares GSM8K free-rollout accuracy against effective k under confidence-adaptive and fixed-k decoding. Both drifting formulations generally improve the accuracy–acceleration trade-off relative to token-only MTP over much of the evaluated range. The tied self-negative formulation remains competitive with conventional drifting and achieves higher accuracy at several intermediate operating points, although neither method consistently dominates. SmoothL1 also remains competitive, particularly at selected operating points. These preliminary results suggest that correcting the self-negative

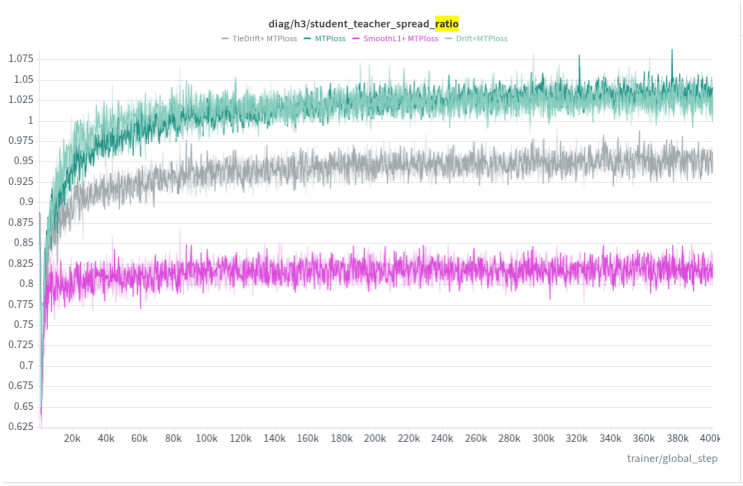


Figure 1: **Effect of latent regularization on relative feature spread at prediction horizon $h = 3$.** The horizontal axis represents the global training step, while the vertical axis shows the ratio of the student's mean pairwise latent-state feature distance to that of the frozen teacher. A ratio of 1 indicates equal average feature spread, whereas values below 1 indicate contraction of student representations relative to the teacher. Paired SmoothL1 regression produces the strongest contraction (approximately 0.82), while conventional drifting and the token-only baseline maintain ratios close to unity (approximately 1.03). The tied self-negative formulation achieves an intermediate ratio (approximately 0.95), preserving more feature dispersion than SmoothL1 while exhibiting slightly greater contraction than conventional drifting. The experimental configurations corresponding to each legend entry are defined in Table 1.

interaction can improve short-horizon representation alignment without sacrificing the generation benefits associated with population-level regularization.

These pilots establish a concrete next step: determine which geometric properties are predictive of rollout quality, and where pointwise regression, drifting, or their combination is preferable. The 8B study will repeat the comparisons with matched budgets, broader bandwidth and horizon sweeps, and stronger validation before moving to the final larger-model experiment.

5 Risks

The wrong neighbours can give the wrong drift. A teacher feature may be close in Euclidean distance without representing a comparable prediction context. Small or heterogeneous pools could therefore push the student in unhelpful directions, especially at longer horizons. We will inspect pool composition and kernel weights by horizon, vary bandwidth and pool size, and omit updates when too few usable neighbours are available.

A better-looking feature space may not generate better answers. The pilot already shows why feature distances cannot be the sole success criterion: SmoothL1 and drifting change spread and transport cost differently. Strong regularization may also damage the token objective. We will compare methods at similar auxiliary gradient strengths and judge improvements primarily by free-rollout accuracy at matched effective k, followed by measured throughput and latency.

Training overhead versus inference gains. Drifting introduces additional computational overhead during training through pairwise feature interactions, which may limit its practicality at larger model scales. We will measure training wall-clock time to quantify this additional cost and compare inference latency and token throughput across different training objectives. These measurements will establish whether improvements in generation quality and inference efficiency justify the additional training cost.ference gain.

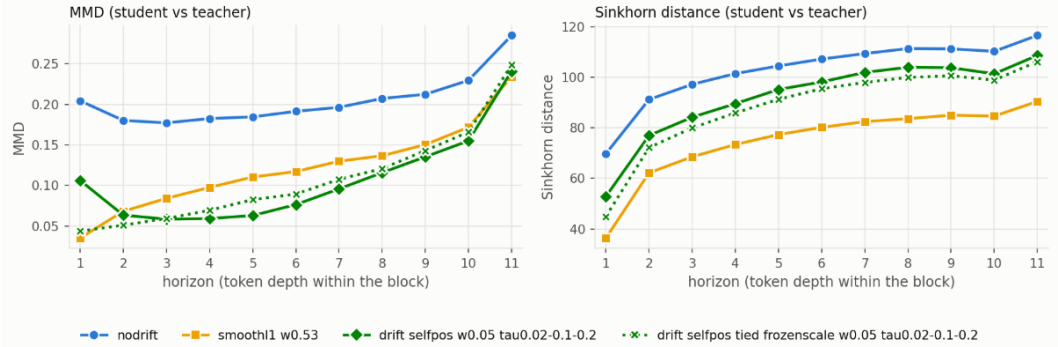


Figure 2: **Tied self-negative drifting improves short-horizon distribution alignment.** Maximum mean discrepancy (MMD; left) and Sinkhorn transport cost (right) between student and frozen-teacher feature distributions across prediction horizons. Lower values indicate closer agreement under the respective metrics. Conventional drifting improves MMD relative to token-only MTP at longer horizons but exhibits a larger discrepancy at $k = 1$. The tied self-negative correction reduces this short-horizon MMD while maintaining comparable performance at longer horizons. SmoothL1 achieves lower Sinkhorn costs across the evaluated horizons, highlighting differences between distributional matching criteria.

Results on one maths benchmark may not transfer. The current evidence comes from MetaMathQA-trained models evaluated on GSM8K. It does not establish benefits for other reasoning tasks or open-ended generation. Larger-model evaluation will therefore include held-out tasks beyond GSM8K, with benchmark overlap checked and model-selection data kept separate from final tests.

6 Timeline & Resource Requirements

The baseline reproduction is complete, and the current work is moving from 1B pilot diagnostics to controlled hyperparameter selection and larger-model validation. The remaining phases depend on compute availability.

1. **Completed:** reproduce on-policy MTP, establish the frozen-teacher reference, and validate free-rollout evaluation.
2. **Ongoing:** compare drifting, SmoothL1, and combined losses at 1B; examine feature geometry and the effects of kernel bandwidth and training horizon.
3. **Next:** use an 8B model to select hyperparameters under matched training budgets, with repeated evaluations and additional reasoning benchmarks.
4. **Final study:** transfer the selected settings to a 32B mixture-of-experts (MoE) model and measure accuracy, effective k, throughput, and latency.

Computing resource request. For 2 billion training tokens and a frozen teacher, we estimate approximately 10 hours of idealized compute for an 8B dense model using four NVIDIA B200 GPUs at 40% model FLOPs utilization. We request access to four B200 GPUs for approximately two days per 8B configuration to allow for MTP-specific overhead, evaluation, and profiling. Multiple configurations and seeds would require additional allocations. The final 32B MoE run will be budgeted after profiling its active-parameter compute and full-model memory requirements. Access to GPUs, storage, and large-model training support from the University of Oxford Department of Computer Science and the MediaTek Computing Research Centre would help complete the proposed comparisons.

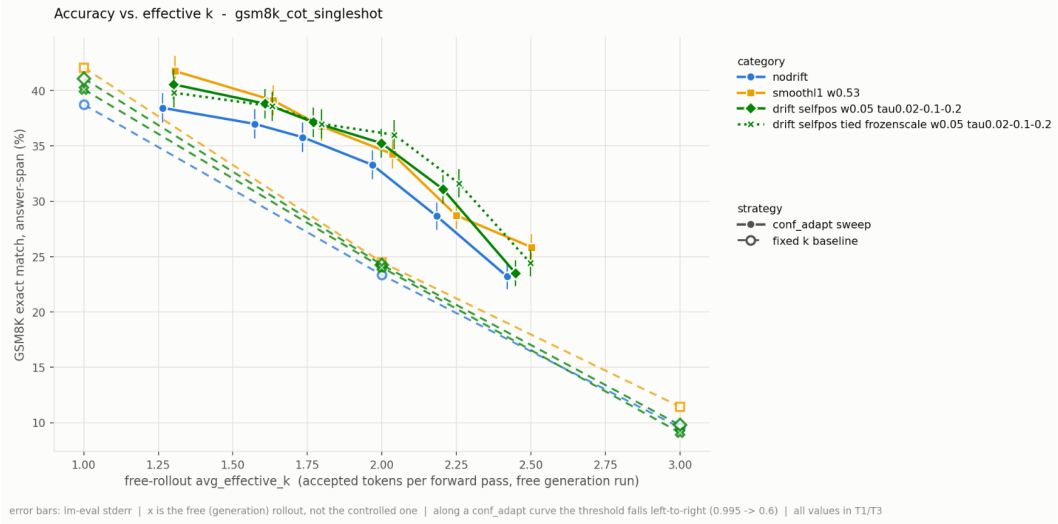


Figure 3: **Accuracy–acceleration trade-off under different latent regularization objectives.** GSM8K free-rollout exact-match accuracy versus the effective number of emitted tokens per forward pass for the 1B pilot models. Filled markers connected by solid lines represent confidence-adaptive decoding, while open markers and dashed lines indicate fixed-(k) decoding. Both conventional and tied self-negative drifting generally improve accuracy over token-only MTP at comparable effective prediction horizons. Tied drifting remains competitive with conventional drifting, with improvements at several intermediate operating points. Error bars indicate evaluation standard errors. Higher accuracy and larger effective (k) are preferable, although effective (k) does not directly measure wall-clock acceleration.

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7 Notation Reference

Table 2: Summary of notation used in the methodology.

Symbol	Definition
$i = (r, j)$	Active prediction position in region r, horizon j
r	MTP prediction region index
j	Prediction horizon within an MTP region
N_{act}	Number of active prediction positions
d	Dimension of the latent feature space
S, T	Student and frozen teacher models
u_i^S, u_i^T	Student and teacher hidden states at position i
ϕ_S, ϕ_T	Student and teacher feature normalization maps
x_i	Normalized student latent representation
y_i^+	Corresponding normalized teacher representation
y_l^-	Detached student representation used as a negative
$\text{sg}(\cdot)$	Stop-gradient operation
$\mathcal{J}_i^+$	Positive (teacher) sample pool for anchor i
$\mathcal{J}_i^-$	Negative (student) sample pool for anchor i
κ_h	Laplace affinity kernel in feature space
τ_h	Kernel temperature (distance scale)
a_{il}^+	Normalized positive affinity weight
a_{il}^-	Normalized negative affinity weight
$\widehat{V}_i$	Estimated drifting field at anchor x_i
η	Drift target displacement scale
ϵ	Numerical stability constant
$\mathcal{L}_{\text{SmoothL1}}$	Paired latent-state regression loss
ρ_β	SmoothL1 penalty function
β	SmoothL1 quadratic-to-linear transition threshold
$\mathcal{L}_{\text{drift}}$	Drifting regularization loss